

JARVIS

Software Requirements Document

Trevor Bank Stock Trading Application

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Introduction

Purpose

The purpose of this document is to describe the requirements and functionalities for the Trevor Bank trading application.

The application would enable existing bank customers to buy and sell stocks through a simple and efficient user-friendly platform. The system aims to simplify stock trading and allow customers to manage investments through their linked bank accounts.

Scope

Trevor Bank plans to develop a stock trading application for retail investors who are existing bank customers.

The system will allow customers to:

- Register by linking their Trevor Bank account
- Log in securely
- View stock market prices
- Buy stocks
- Sell stocks
- Track their investment portfolio
- View transaction history

The platform focuses on simplicity and accessibility for beginner investors.

Out of Scope

The following features are not included in this system:

- Cryptocurrency trading
- Options trading
- Margin trading
- Alert Systems

System Overview

Product Perspective

The Trevor Bank Stock Trading Application will be a full-stack financial platform consisting of:

- Mobile and web user interfaces
- Backend trading services
- Integration with Trevor Bank's customer database
- Integration with stock market data providers
- Database for storing users, trades, and portfolios

The system will allow users to link their bank accounts to trade stocks.

Product Objectives

The objectives of the application are:

- Provide a simple trading platform for beginner investors
- Enable customers to invest directly using their Trevor Bank accounts
- Offer real-time stock price information
- Allow users to track portfolio performance
- Ensure secure financial transactions

User Classes

- Retail Investor : Existing customers who'll use the app to trade stocks
- Admin: Bank employees responsible for monitoring accounts and transactions.

Functional Requirements

This section describes the functions of features of a system. The following discussed are the requirements that define the services that a system must deliver.

Authentication (Login)

Description: User can log in using bank account number and password.

Acceptance criteria:

- Valid credentials authenticate successfully.
- Invalid credentials show error without revealing which field is wrong.
- Session created with timeout rules.

Bank Account Linking / Association

Description: After login, the app associates the user with their Trevor Bank account(s) used for funding/trading.

Acceptance criteria:

- User sees which account is linked (masked identifiers).

Multi-Factor Authentication (MFA) with Biometrics

Description: Support MFA, including biometric authentication on supported devices.

Acceptance criteria:

- User can enable/complete MFA during login.
- Fallback MFA method exists if biometrics unavailable.

App Navigation / Tabs

Description: App provides simple navigation with the following tabs/screens:

- Trade
- Stock Info
- Overview (Reports)
- Positions

Acceptance criteria:

- Tabs are accessible and consistent.
- Users can complete core workflows in minimal steps.

Stock Related Information

Description: User can search/select a stock and view basic performance information.

Includes:

- Ticker and company name
- Current price
- Daily change (\$ and %)
- Basic historical performance view (simple)

Acceptance criteria:

- Search returns relevant results.
- Stock detail page loads within performance targets.

Place Buy Order

Description: User can place a simple Buy order.

Minimum fields:

- Account (preselected)
- Stock (selected)
- Quantity
- Review screen
- Submit

Acceptance criteria:

- Order submission returns confirmation.
- User can see order reflected in positions (immediate).

Place Sell Order

Description: User can sell from owned positions.

Acceptance criteria:

- Only owned shares can be sold .
- Confirmation displayed.

- Position updates after execution/settlement rules.

Positions / Portfolio View

Description: User can view current holdings and basic metrics.

Minimum display fields:

- Stock/ticker
- Quantity
- Date of purchase
- Current value
- Profit/Loss (P/L) (%)

Acceptance criteria:

- Portfolio list loads reliably.
- P/L calculations are consistent with defined formula.

Overview Reports

Description: Provide a simple overview/report screen.

Minimum:

- Total portfolio value
- Daily change
- Basic summary of gains/losses
- Recent activity list (buy/sell history)

Acceptance criteria:

Clear “at a glance” view for non-tech-savvy users.

Trade / Activity History

Description: User can view trade history (buy/sell events).

Minimum fields:

- Timestamp

- Stock
- Buy/Sell
- Quantity
- Price (if available)

Acceptance criteria:

History is searchable/filterable.

Data Collection

Description: App collects usage and financial/trade activity data (within privacy limits).

Fields:

- Usage events (screens visited, session duration, drop-off)
- Trade events (buy/sell, amounts spent, number of stocks purchased)
- Access logs (who accessed, how often)

Acceptance criteria:

- Events are logged with anonymization/pseudonymization where required.
- Data stored/processed on AWS per architecture.

Accessibility

Description: App must be screen-reader capable and support basic accessibility needs.

Acceptance criteria:

- Key flows (login, browse, trade, positions) usable with screen readers.
- Proper labels, focus order, and contrast.

Non-functional Requirements

The following are the requirements that describe how a system should perform. These are discussed below.

Performance (latency)

- Trade submission should complete within 0.1 seconds .
- Stock/portfolio screens should load fast enough for smooth usage.

Scalability / Concurrency

Expected concurrent usage: 200,000 users at the same time

Acceptance criteria

- System supports 200k concurrent sessions.
- No data corruption under load.

Security

- MFA required (with biometrics)
- Secure session handling (timeouts, device binding if needed)

Privacy & Compliance

- Collect usage and trading data while respecting privacy and internal policies.
- Access logging (“who’s accessing/how often”) must be controlled.

Availability & Reliability

- Uptime target for PoC environment — To be decided
- Error rate threshold — To be decided

Observability

Logging, metrics, tracing for:

- login failures
- trade submission outcomes
- latency
- dependency failures (market data/trading API)

Maintainability

- Clean modular architecture suitable for iterative expansion beyond PoC
- Clear API contracts + documentation

Data model

This PoC will store only the minimum data required to support trading operations.

Customer and bank account information already exists in the Trevor Bank core banking system, therefore the trading application database will reference those records using their unique identifiers.

The following tables represent the basic data entities required for the application.

External Entities

Owned by Trevor Bank Core Systems

The following entities are maintained in Trevor Bank's existing systems. The trading application will not store these records; it will only reference them using the identifiers listed below.

Customer

Field (Referenced by Trading App)	Description
customer_id (PK)	Unique identifier for the customer (from bank system)

The trading application will not store customer profile attributes (e.g., name, email) in its database for the PoC.

BankAccount

Field (Referenced by Trading App)	Description
bank_account_id (PK)	Unique identifier for the bank account
customer_id (FK)	Links the account to the owning customer

Internal Entities

Trading Application Entities

TradingAccount

Purpose: Represents the trading profile and the linked funding account used for trading.

Field	Description
trading_account_id (PK)	Unique identifier for the trading account (internal)
customer_id (FK To Customer.customer_id)	Customer who owns this trading account
bank_account_id (FK To BankAccount.bank_account_id, UNIQUE)	Funding account linked for trading
created_at	When the trading account link was created

Stock (Instrument)

Purpose: Stores minimum stock reference data used by the app (not live prices).

Field	Description
stock_id (PK)	Unique identifier for the stock (internal)
ticker	Stock ticker symbol (e.g., AAPL)
company_name	Company name

Price/performance data can be fetched from an external market data provider; this table is just a reference list for IDs, tickers, and display name.

Order

Purpose: Stores buy/sell requests submitted by the user.

Field	Description
order_id (PK)	Unique order identifier
trading_account_id (FK → TradingAccount.trading_account_id)	Trading account placing the order
stock_id (FK → Stock.stock_id)	Stock being traded
side	BUY or SELL
quantity	Number of shares
status	SUBMITTED / COMPLETED / REJECTED
order_date	Date/time the order was placed

Position

Purpose: Stores current holdings per trading account and stock.

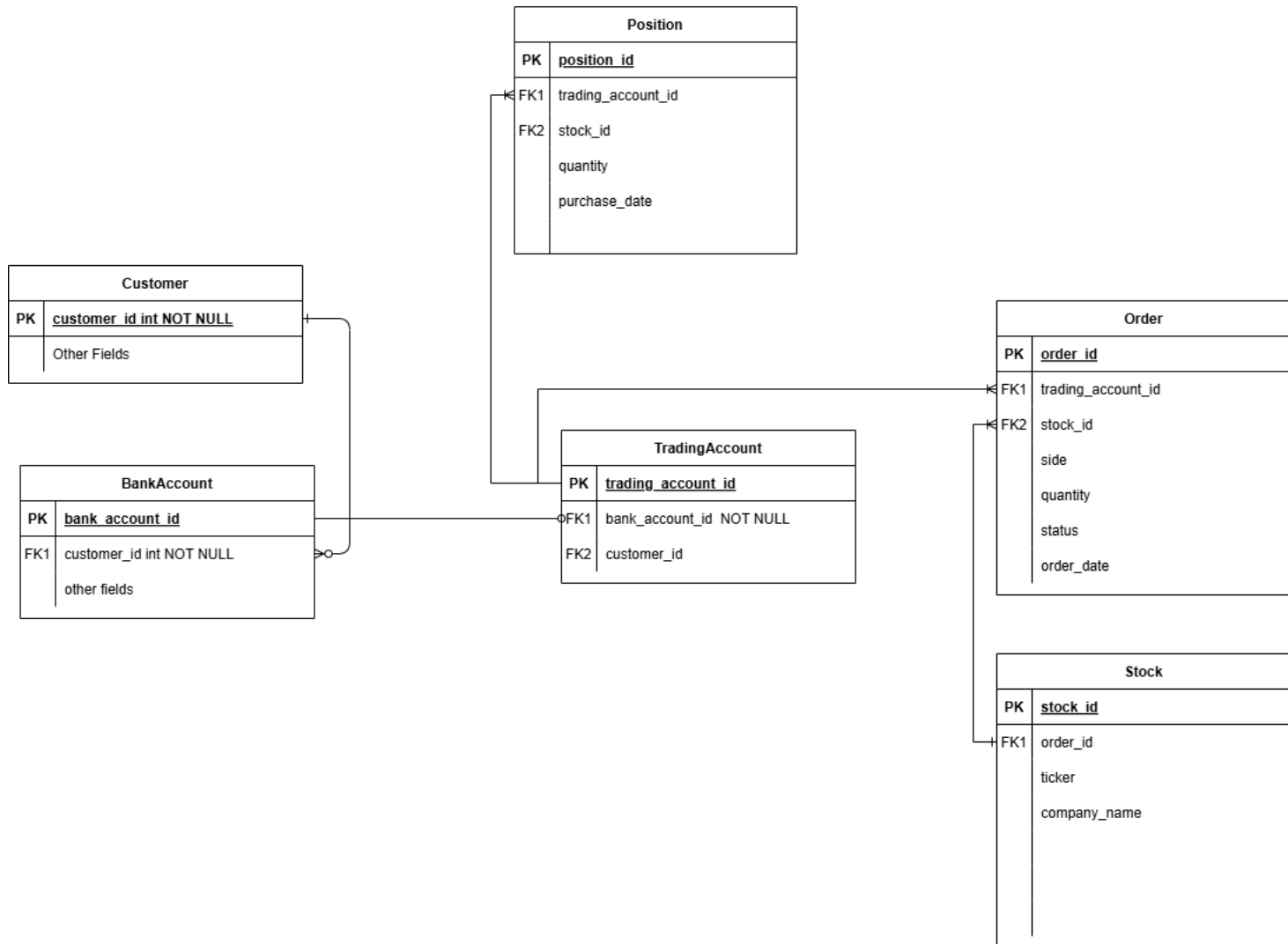
Field	Description
position_id (PK)	Unique position identifier
trading_account_id (FK to TradingAccount.trading_account_id)	Trading account holding the stock
stock_id (FK to Stock.stock_id)	Stock held
quantity	Shares currently held
purchase_date	Date of purchase

Relationship Summary

- One Customer can have one or more Bank Accounts
- One Bank Account can have only one Trading Account
- One Trading Account can place many Orders
- One Trading Account can hold many Positions
- One Stock can appear in many Orders
- One Stock can appear in many Positions

Entity Relationship Diagram

ERD FOR TRADING APP



Dependencies

The following external systems and services are required for the proper functioning of the Trevor Bank Retail Stock Trading Application.

1. Trevor Bank Core Banking System

The application depends on the bank's core system to retrieve customer information and bank account details. User authentication and account verification rely on this system.

2. Authentication and Security Services

The system depends on Trevor Bank's authentication infrastructure for login validation, password verification, and multi-factor authentication (MFA) including biometric authentication.

3. Stock Market Data Provider

The application requires integration with an external market data provider to retrieve stock information such as ticker symbols, price data, and performance metrics.

4. Trade Execution System / Brokerage Service

The application depends on a trade execution platform or brokerage system that processes buy and sell orders placed by users.

5. Cloud Infrastructure (AWS)

The system depends on Amazon Web Services (AWS) for data storage, processing, and application hosting.

Assumptions and constraints

System Constraints

- Only Trevor Bank customers can use the system.

- Only stock trading is supported.
- Only market orders are supported in the first version.
- The system must integrate with bank infrastructure.

Assumptions

- It is assumed that customers using the application are already authorized and eligible to trade stocks through Trevor Bank (no KYC/onboarding flows are included in this PoC).
- It is assumed that a trade execution system/brokerage service exists and is available to process buy and sell orders submitted by the application.
- It is assumed that a reliable source of stock market data is available (external provider or internal feed) to supply stock prices and basic performance information.

Use Cases

The following table summarizes the primary user interactions supported by the Trevor Bank Stock Trading Application.

Use Case	Description
Register Account	• User enters bank account number and password. • System verifies the user with Trevor Bank database. • User completes biometric MFA. • Account is successfully linked to the trading app.
Login	• User logs in using account number and password. • System validates credentials. • User completes biometric authentication. • User is directed to the Overview

	dashboard.
Buy Stock	• User navigates to the Trade tab. • User selects a stock and enters quantity. • System shows order summary for confirmation. • Trade is executed and portfolio is updated.
Sell Stock	• User opens the Positions tab. • User selects stock to sell and enters quantity. • System shows confirmation details. • Trade is executed and transaction history is updated.

Future Scope

Possible future features include:

- Cryptocurrency trading
- Options trading
- AI-based investment insights
- Automated portfolio management